

HDOC DAY-17

Digit-Based Operations Using Switch-Case and User-Defined Functions

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Aim

To write a menu-driven C program using **switch-case** and separate **void user-defined functions** to perform different digit-based operations on a positive integer.

Problem Statement

Write a menu-driven C program using **switch-case**. Define one separate user-defined function for each menu option, with **void return type** and a simple function call. Each function should read the required input, process it, and display the result.

1. Generate the largest possible number using its digits.
2. Generate the smallest possible number using its digits.
3. Swap its first and last digits.
4. Check whether it is a palindrome number.

Algorithm

1. Start the program and display four menu options.
2. Read the user choice in **main()**.
3. Use **switch-case** to call the selected function.
4. In **largest()**, count every digit and print digits from 9 to 0.
5. In **smallest()**, count every digit; print the smallest non-zero digit first, then zeros, then the remaining digits in ascending order.
6. In **swapDigits()**, find the first digit, last digit and place value, then interchange the first and last digits.
7. In **palindrome()**, reverse the number and compare the reversed number with the original number.
8. Display the result and stop.

Test Case Table

Choice	Input	Operation	Expected Output
1	42153	Largest possible number	54321
1	50291	Largest possible number	95210
2	42153	Smallest possible number	12345
2	50291	Smallest possible number	10259
3	12345	Swap first and last digits	52341
3	9876	Swap first and last digits	6879
4	12321	Palindrome check	Palindrome
4	12345	Palindrome check	Not palindrome

C Program

```
#include <stdio.h>

void largest();
void smallest();
void swapDigits();
void palindrome();

int main()
{
    int choice;

    printf("\nMENU\n");
    printf("1. Generate largest possible number\n");
    printf("2. Generate smallest possible number\n");
    printf("3. Swap first and last digits\n");
    printf("4. Check palindrome number\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch(choice)
    {
        case 1: largest(); break;
        case 2: smallest(); break;
        case 3: swapDigits(); break;
        case 4: palindrome(); break;
        default: printf("Invalid choice\n");
    }
    return 0;
}

void largest()
{
    int n, digit, count[10] = {0}, i;
    printf("Enter a positive integer: ");
    scanf("%d", &n);

    while(n > 0) {
        digit = n % 10;
        count[digit]++;
        n = n / 10;
    }

    printf("Largest number = ");
    for(i = 9; i >= 0; i--)
        while(count[i] > 0) {
            printf("%d", i);
            count[i]--;
        }
    printf("\n");
}

void smallest()
{
    int n, digit, count[10] = {0}, i;
    printf("Enter a positive integer: ");
    scanf("%d", &n);

    while(n > 0) {
        digit = n % 10;
        count[digit]++;
        n = n / 10;
    }

    printf("Smallest number = ");
    for(i = 1; i <= 9; i++) {
        if(count[i] > 0) {
            printf("%d", i);
            count[i]--;
            break;
        }
    }
}
```

```

while(count[0] > 0) {
    printf("0");
    count[0]--;
}
for(i = 1; i <= 9; i++)
    while(count[i] > 0) {
        printf("%d", i);
        count[i]--;
    }
printf("\n");
}

void swapDigits()
{
    int n, first, last, temp, place = 1, result;
    printf("Enter a positive integer: ");
    scanf("%d", &n);

    last = n % 10;
    temp = n;
    while(temp >= 10) {
        temp = temp / 10;
        place = place * 10;
    }
    first = temp;
    result = n - first * place - last;
    result = result + last * place + first;
    printf("Number after swapping = %d\n", result);
}

void palindrome()
{
    int n, temp, reverse = 0, digit;
    printf("Enter a positive integer: ");
    scanf("%d", &n);
    temp = n;

    while(temp > 0) {
        digit = temp % 10;
        reverse = reverse * 10 + digit;
        temp = temp / 10;
    }

    if(reverse == n)
        printf("%d is a palindrome number\n", n);
    else
        printf("%d is not a palindrome number\n", n);
}

```

Compilation and Execution

The program can be compiled using a standard C compiler. Example:

```
gcc hdoc17.c -o hdoc17
./hdoc17
```

Sample Outputs

Test 1 - Largest Number

```
MENU
1. Generate largest possible number
2. Generate smallest possible number
3. Swap first and last digits
4. Check palindrome number
Enter your choice: 1
Enter a positive integer: 42153
Largest number = 54321
```

Test 2 - Smallest Number

```
Enter your choice: 2
Enter a positive integer: 50291
Smallest number = 10259
```

Test 3 - Swap Digits

```
Enter your choice: 3
Enter a positive integer: 12345
Number after swapping = 52341
```

Test 4 - Palindrome

```
Enter your choice: 4
Enter a positive integer: 12321
12321 is a palindrome number
```

Result

The menu-driven C program was written and tested successfully. All four digit-based operations produced the expected results using separate **void** user-defined functions and **switch-case**.